

Exercise 4.1 *It is possible to construct distributions that have a large gap between min-entropy and Shannon entropy. This shows that controlling the Shannon entropy is not sufficient for some cryptographic tasks.*

- a.) Given $\epsilon \in (0, 1)$, construct a sequence of random variables $(X_1, X_2, \dots, X_n, \dots)$ where $X_n \in \{0, 1, \dots, n\}$, such that

$$\left. \begin{aligned} H(X_n) &\geq (1 - \epsilon) \log n \\ H_{\min}(X_n) &= C, \end{aligned} \right\} \forall n \geq N$$

for some $N \in \mathbb{N}$ and some constant $C > 0$.

- b.) Given $\epsilon \in (0, 1)$, construct a sequence of random variables $(X_1, Y_1), (X_2, Y_2), \dots$, where $X_n, Y_n \in \{0, 1, \dots, n\}$, such that

$$\left. \begin{aligned} H(X_n) &= H_{\min}(X_n) = \log n \\ H(X_n|Y_n) &\geq (1 - \epsilon) \log n \\ H_{\min}(X_n|Y_n) &= C \end{aligned} \right\} \forall n \geq N$$

for some $N \in \mathbb{N}$ and some constant $C > 0$.

Solution:

- a.) For each $n = 1, 3, 4, \dots$, we consider the following distribution

$$P_{X_n}(x) = \begin{cases} \epsilon & \text{if } x = 0, \\ \frac{1-\epsilon}{n-1} & \text{otherwise.} \end{cases}$$

In this case,

$$\begin{aligned} H(X_n) &= H(\epsilon) + (1 - \epsilon) \cdot \log(n - 1) \geq (1 - \epsilon) \cdot \log n & \forall n \geq \frac{1 - \epsilon}{H(\epsilon)} + 1, \\ H_{\min}(X_n) &= -\log \epsilon & \forall n \geq \frac{1 - \epsilon}{\epsilon} + 1. \end{aligned}$$

Thus the construction satisfies the requirements by letting $N = \left\lceil \max\left\{\frac{1-\epsilon}{H(\epsilon)} + 1, \frac{1-\epsilon}{\epsilon} + 1\right\} \right\rceil$, and $C = -\log \epsilon$.

- b.) For each $n = 1, 2, 3, 4, \dots$, we consider the following distribution

$$\begin{aligned} P_{Y_n}(y) &= \frac{1}{n} \\ P_{X_n|Y_n}(x|y) &= \begin{cases} \epsilon & \text{if } x = y, \\ \frac{1-\epsilon}{n-1} & \text{otherwise.} \end{cases} \end{aligned}$$

In this case, $P_{X_n}(x) = \sum_y P_{Y_n}(y) \cdot P_{X_n|Y_n}(x|y) = 1/n$. Thus, $H(X_n) = H_{\min} = \log n$. Additionally,

$$\begin{aligned} H(X_n|Y_n) &= H(\epsilon) + (1 - \epsilon) \cdot \log(n - 1) \geq (1 - \epsilon) \cdot \log n & \forall n \geq \frac{1 - \epsilon}{H(\epsilon)} + 1, \\ H_{\min}(X_n|Y_n) &= -\log \epsilon & \forall n \geq \frac{1 - \epsilon}{\epsilon} + 1. \end{aligned}$$

Thus the construction satisfies the requirements by letting $N = \left\lceil \max\left\{\frac{1-\epsilon}{H(\epsilon)} + 1, \frac{1-\epsilon}{\epsilon} + 1\right\} \right\rceil$, and $C = -\log \epsilon$.

Exercise 4.2 (Rényi entropy) Both the min-entropy and the Shannon entropy are limiting cases of the following family of Rényi entropies:

$$H_\alpha(X) = \frac{1}{1-\alpha} \log \sum_x P(x)^\alpha, \quad \alpha \in (0, 1) \cup (1, +\infty).$$

- a) Compute the limit of the above quantities for $\alpha \rightarrow \{0, 1, +\infty\}$.
b) Plot the Rényi entropy as a function of α for the random variable X distributed as

x	0	1	2
$P(x)$	1/2	1/4	1/4

- c) Show that, for any random variable $X \in \mathcal{X}$ and any pmf $P(x)$, the Rényi entropy is monotonically non-increasing in the parameter α . This yields an alternative proof of the fact that $H_{\min}(X) \leq H(X) \leq \log |\mathcal{X}|$.

Hint: Compute $\frac{d}{d\alpha} H_\alpha$ and then express it in the form of $f(\mathbb{E}[t]) - \mathbb{E}[f(t)]$, where $f(t) = t \log t$ and $t = P(x)^{\alpha-1}$.

Solution:

- a.) We obtain

$$\lim_{\alpha \rightarrow 0+} H_\alpha(X) = \log \left(\sum_x \lim_{\alpha \rightarrow 0+} P(x)^\alpha \right) = \log |\{x : P(x) > 0\}| = H_{\max}(X),$$

and

$$\begin{aligned} \lim_{\alpha \rightarrow 1} H_\alpha(X) &= \frac{(\sum_x P(x))^{-1} \cdot \sum_x \log P(x) \cdot P(x)^\alpha}{-1} \Big|_{\alpha=1} \\ &= - \sum_x P(x) \cdot \log P(x) = H(X), \end{aligned}$$

where we have used L'Hôpital's rule, and

$$\begin{aligned} \lim_{\alpha \rightarrow +\infty} H_\alpha(X) &= - \lim_{\alpha \rightarrow +\infty} \log \left[\left(\sum_x P_{\max}^\alpha \cdot \left(\frac{P(x)}{P_{\max}} \right)^\alpha \right)^{\frac{1}{\alpha-1}} \right] \\ &= - \lim_{\alpha \rightarrow +\infty} \frac{\alpha}{\alpha-1} \cdot \log P_{\max} + \frac{1}{\alpha-1} \cdot \log \left(\sum_x \left(\frac{P(x)}{P_{\max}} \right)^\alpha \right) \\ &= - \log P_{\max} = H_{\min}(X), \end{aligned}$$

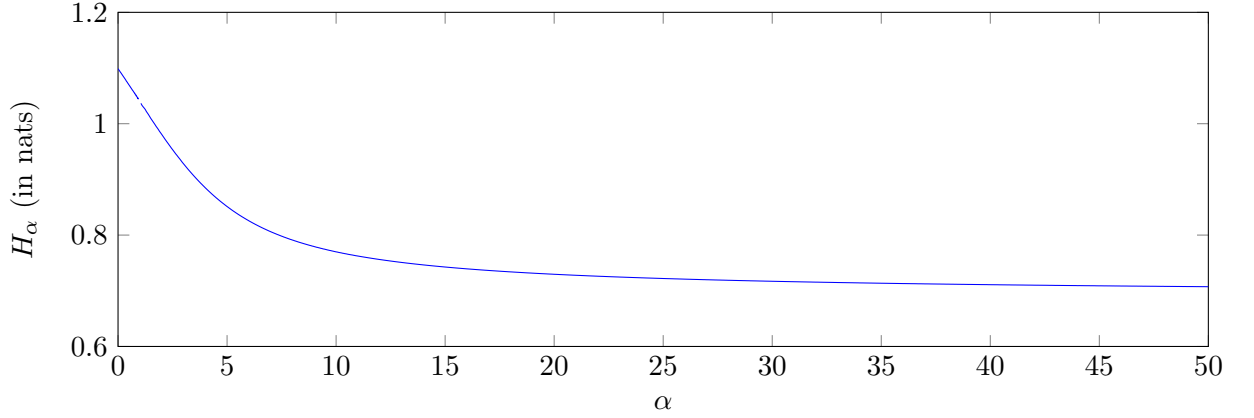
where we have defined

$$P_{\max} \triangleq \max_x P(x),$$

and used

$$1 \leq \sum_x \left(\frac{P(x)}{P_{\max}} \right)^\alpha \leq |\mathcal{X}|.$$

- b.) We plot the Rényi entropy of the above random variable in nats.



c.) Consider the derivative of H_α w.r.t. α

$$\frac{d}{d\alpha} H_\alpha(X) = \frac{(1-\alpha) \sum_{x'} P(x')^\alpha \log P(x') + \sum_x P(x)^\alpha \log \sum_{x'} P(x')^\alpha}{(1-\alpha)^2 \sum_x P(x)^\alpha}.$$

Note that $\lim_{\alpha \rightarrow 1} H_\alpha$ exists from both sides. It suffices to show $\frac{d}{d\alpha} H_\alpha(X)$ to be non-positive for all $\alpha \in (0, 1) \cup (1, \infty)$.

Let

$$F(\alpha) = (1-\alpha) \sum_{x'} P(x')^\alpha \log P(x') + \sum_x P(x)^\alpha \log \sum_{x'} P(x')^\alpha.$$

It is sufficient to show that F is non-positive for $\alpha > 0$.

Note that

$$\begin{aligned} \sum_x P(x)^\alpha &= \sum_x P(x) P(x)^{\alpha-1} = \mathbb{E}[P(x)^{\alpha-1}], \\ \sum_x P(x)^\alpha \log P(x) &= \frac{\sum_x P(x) P(x)^{\alpha-1} \log P(x)^{\alpha-1}}{\alpha-1} = \frac{\mathbb{E}[P(x)^{\alpha-1} \log P(x)^{\alpha-1}]}{\alpha-1}. \end{aligned}$$

We also denote $f(t) = t \log t$. Recall that $f(t)$ is convex on $(0, 1)$. Therefore, we obtain

$$F(\alpha) = f(\mathbb{E}[P(x)^{\alpha-1}]) - \mathbb{E}[f(P(x)^{\alpha-1})] \leq 0,$$

which proves the non-positivity of $F(\alpha)$.

Exercise 4.3 Consider a source (X, Y) where Y is a uniformly distributed bit and X follows the following conditional distribution:

$P_{X Y}$	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$
$y = 1$	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{2}$	0
$y = 2$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	0	$\frac{1}{2}$

- Find an extractor, a function $f : x \mapsto f(x) \in \{0, 1\}$, such that $Z = f(X)$ is uniformly distributed and independent of Y .
- Compute the marginal distribution P_X , as well as the entropy $H_{\min}(X)$.
- Compute the conditional entropy $H_{\min}(X|Y)$.
- Compare $H_{\min}(X)$ with $H_{\min}(X|Y)$.

- Why is this relationship between the two quantities expected?

- Is the extractor in the first item optimal?
- Could there be a better extractor if we only require Z to be uniformly distributed, but not necessarily independent of Y ?

Solution:

- a.) The asked extractor can be the following function $f : x \mapsto 1$ for $x = 4, 5$, and $x \mapsto 0$ for $x = 1, 2, 3$.
- b.) $P_X = (3/16, 1/8, 3/16, 1/4, 1/4)$ and $H_{\min}(X) = 2$.
- c.)

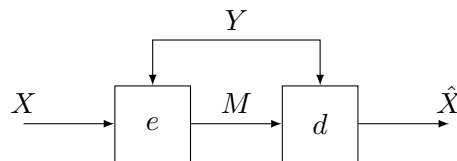
$$\begin{aligned}
 H_{\min}(X|Y) &= -\log \left(\sum_{y=1,2} P_Y(y) \cdot \max_{x \in \{1, \dots, 5\}} P_{X|Y}(x|y) \right) \\
 &= -\log \left(\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \right) = 1 \text{ bit}, \\
 H(X|Y) &= - \sum_{y=1,2, x \in \{1, \dots, 5\}} P_{XY}(x, y) \cdot \log P_{X|Y}(x|y) \\
 &= \dots = \frac{13}{8} \text{ bits.}
 \end{aligned}$$

- d.) The entropy with side information can never be larger than the entropy without side information because of data-processing, so we expect $H_{\min}(X) \geq H_{\min}(X|Y)$. We have $H_{\min}(X|Y) = 1$, so extracting one bit is already optimal if we require independence from Y . Since we cannot partition P_X into four subsets of equal weight, extracting more than one bit from P_X is not possible even though $H_{\min}(X) = 2$.

Exercise 4.4 Consider a source producing random variables X and Y with the following joint distribution:

P_{XY}	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$
$y = 1$	$\frac{1}{32}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{1}{32}$
$y = 2$	$\frac{1}{8}$	0	$\frac{1}{8}$	$\frac{1}{4}$	0

- a) Compute $H_{\min}(X|Y)$ and $H(X|Y)$.
- b) Assume now that $Y = y$ has been observed. Find an optimal source code for $P_{X|Y=y}$ for $y \in \{1, 2\}$.
- c) We are interested in compressing X losslessly. Assume that both the encoder and the decoder have access to Y (see the diagram below). Find a variable-length source code for this setting, i.e., an encoder $e : \mathcal{X} \times \mathcal{Y} \rightarrow \{0, 1\}^*$ and a decoder $d : \{0, 1\}^* \times \mathcal{Y} \rightarrow \mathcal{X}$.



- d) What is the expected codeword length of this code? (Note that you need to take the expectation over both Y and X .)

- e) Find an extractor that extracts exactly $H_{\min}(X|Y)$ bits of uniform and independent randomness from X .

Solution:

- a.) We calculate

$$\begin{aligned}
 H_{\min}(X|Y) &= -\log \left(\sum_{y=1,2} P_Y(y) \cdot \max_{x \in \{1,\dots,5\}} P_{X|Y}(x|y) \right) \\
 &= -\log \left(\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \right) = 1 \text{ bit}, \\
 H(X|Y) &= - \sum_{y=1,2, x \in \{1,\dots,5\}} P_{XY}(x, y) \cdot \log P_{X|Y}(x|y) \\
 &= - \left(\frac{2}{32} \log \frac{1}{16} + \frac{1}{16} \log \frac{1}{8} + \frac{3}{8} \log \frac{1}{4} + \frac{2}{4} \log \frac{1}{2} \right) = \frac{27}{16} \text{ bits}.
 \end{aligned}$$

- b.) Since the conditional probabilities are of the form $2^{-\ell}$,

$P_{X Y}$	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$
$y = 1$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{16}$
$y = 2$	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{2}$	0

we know that the Shannon code is optimal,

P_{XY}	$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$
$y = 1$	1110	10	110	0	1111
$y = 2$	10	—	11	0	—

- c.) We use the code we obtained in the last step for the encoder. Namely, depending on $Y = 0$ or $Y = 1$, we encoding and accordingly decode using one of the two Shannon codes.
- d.) $\bar{l} = 2\frac{1}{32}4 + 1\frac{1}{16}3 + 3\frac{1}{8}2 + 2\frac{1}{4}1 = 27/16$
- e.) This could be $f : x \mapsto 1$ for $x \in \{4, 5\}$, and $x \mapsto 0$ for $x \in \{1, 2, 3\}$.

Exercise 4.5 (AEP for smooth min-entropy) We are interested in showing that $\frac{1}{n}H_{\min}^{\epsilon}(X^n) \rightarrow H(X)$ as $n \rightarrow \infty$ and $\epsilon \rightarrow 0$ (in that order), where X^n is i.i.d. following P_X .

- a) For each $\epsilon > 0$, let $\mathcal{A}_{\epsilon}^{(n)}(X)$ denote the ϵ -typical set of X . For each positive integer n , define a distribution on $\mathcal{X}^n$ as

$$\hat{P}_{X^n}(x^n) := \begin{cases} P_{X^n}(x^n) & x^n \in \mathcal{A}_{\epsilon}^{(n)}(X) \\ \frac{1 - P_{X^n}(\mathcal{A}_{\epsilon}^{(n)}(X))}{|\mathcal{X}|^n - |\mathcal{A}_{\epsilon}^{(n)}(X)|} & x^n \notin \mathcal{A}_{\epsilon}^{(n)}(X) \end{cases}$$

Prove that for n large enough, $\delta_{\text{td}}(P_{X^n}, \hat{P}_{X^n}) \leq \epsilon$.

- b) Using $\hat{P}_{X^n}$ from the previous step, show that

$$\frac{1}{n}H_{\min}^{\epsilon}(X^n) \geq H(X) - \epsilon$$

for n large enough.

c.) Using the continuity bound from Exercise 3.5, show that

$$\frac{1}{n} H_{\min}^{\epsilon}(X^n) \leq H(X) + \frac{1}{n} h(\epsilon) + \epsilon \cdot \log |\mathcal{X}|$$

for $0 < \epsilon < 1/2$.

The desired limits now follow.

Solution:

a.) Note that

$$\begin{aligned} \frac{1}{2} \sum_{\mathbf{x}^n} \left| P_{X^n}(\mathbf{x}^n) - \hat{P}_{X^n}(\mathbf{x}^n) \right| &= \frac{1}{2} \sum_{\mathbf{x}^n \in \mathcal{X}^n \setminus \mathcal{A}_{\epsilon}^{(n)}(X)} \left| P_{X^n}(\mathbf{x}^n) - \hat{P}_{X^n}(\mathbf{x}^n) \right| \\ &\leq \frac{1}{2} \sum_{\mathbf{x}^n \in \mathcal{X}^n \setminus \mathcal{A}_{\epsilon}^{(n)}(X)} P_{X^n}(\mathbf{x}^n) + \hat{P}_{X^n}(\mathbf{x}^n) \\ &= 1 - P_{X^n}(\mathcal{A}_{\epsilon}^{(n)}(X)) \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$ for any fixed $\epsilon > 0$. Thus, there must exist some $N \in \mathbb{N}$ such that $\delta_{\text{tvd}}(P_{X^n}, \hat{P}_{X^n}) \leq \epsilon$ for all $n \geq N$.

b.) If $\mathcal{X}^n \setminus \mathcal{A}_{\epsilon}^{(n)}(X) = \emptyset$,

$$\max_{\mathbf{x}^n} \hat{P}_{X^n}(\mathbf{x}^n) = \max_{\mathbf{x}^n \in \mathcal{A}_{\epsilon}^{(n)}(X)} P_{X^n}(\mathbf{x}^n) \leq 2^{-n(H(X)-\epsilon)}.$$

If $\mathcal{X}^n \setminus \mathcal{A}_{\epsilon}^{(n)}(X) \neq \emptyset$,

$$\frac{1 - P_{X^n}(\mathcal{A}_{\epsilon}^{(n)}(X))}{|\mathcal{X}|^n - |\mathcal{A}_{\epsilon}^{(n)}(X)|} \leq 1 - P_{X^n}(\mathcal{A}_{\epsilon}^{(n)}(X)) \leq 2^{-n(H(X)-\epsilon)}$$

as $n \rightarrow \infty$. Thus, for n large enough,

$$\begin{aligned} \frac{1}{n} H_{\min}^{\epsilon}(X) &= \frac{1}{n} \max_{\delta_{\text{tvd}}(P_X, \tilde{P}_X) \leq \epsilon} H_{\min}(X)_{\tilde{P}_X} \\ &\geq \frac{1}{n} H_{\min}(X)_{\hat{P}_X} \geq -\frac{1}{n} \log 2^{-n(H(X)-\epsilon)} = H(X) - \epsilon. \end{aligned}$$

c.) Suppose $H_{\min}^{\epsilon}(X^n) = H_{\min}(X^n)_{P_{X^n}^*}$, where $\delta_{\text{tvd}}(P_X, P_X^*) \leq \epsilon$. Then,

$$\begin{aligned} \frac{1}{n} H_{\min}^{\epsilon}(X^n) &= \frac{1}{n} H_{\min}(X^n)_{P_{X^n}^*} \leq \frac{1}{n} H(X^n)_{P_{X^n}^*} \leq \frac{1}{n} (H(X^n) + h(\epsilon) + \epsilon \cdot \log |\mathcal{X}|^n) \\ &= H(X) + \frac{1}{n} \cdot (h(\epsilon) + \epsilon \cdot n \log |\mathcal{X}|). \end{aligned}$$